STAFF REVIEW · H446 044 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. Named input to a method

2. A debugger stopping point

3. FIFO versus LIFO

4. NOT B=1; A AND NOT B=1; then 1 OR C=1. Final output: 1.

I Do — source worked steps

1. Identify class, objects and members — Badge is the class (blueprint). amber and blue are references to two objects . Each has its own Points field and can run the Add method.

2. Trace the state change — new Badge() creates each object with Points = 0 . amber.Add(2) changes amber’s field from 0 to 2; blue remains at 0 because it is a different instance.

3. Predict, then check the output — The output is 2,0 . A class definition does not create an object. A method call on one object does not automatically change the fields of another.

We Do — guided CHECK

1. Independent state
first.Value = 2 and second.Value = 0 . Both calls act on first ; they do not change second .

2. Predict before checking
Which instance runs Increase() ? Which instance field changes?
Targeted update
first.Value = 2 ; second.Value = 1 . The variable second refers to the second created Counter instance.

You Do — source mark schemes

1. class Book { public string Title; }

2. Book novel = new Book();

3. Each object has its own instance state.

Exit Ticket — source answer

1. False. A class describes the structure and behaviour of objects, but defining the class does not instantiate anything. An object exists only when code creates an instance, for example using new.